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1.0 Engineering Problem

Modern Buildings rely on a heating, ventilation and air-conditioning (HVAC) system to maintain a comfortable indoor condition for the occupants. However, throughout the day the amount of people in the building can vary significantly affecting the demand of the system to control the conditions of the room. Normally HVAC systems are on fix schedule which may not accurately represent the occupancy of the room or may take longer to change to the rooms new occupancy which can waste energy and is inefficient. Therefore, the use of a numerical system can be incorporated to make the system more responsive by predicting the current occupancy. The current data being provided gives the occupancy of 5 rooms inside the building. However, the models will only be trained and validated for one of the rooms and it is assumed that it can be re-trained against the data in the other rooms and be just as successful. The aim of this project is developing a decision tree regression model capable of predicting room occupancy 15 minutes in advance using historical occupancy measurements and time-based variables. The model will be evaluated against Historical Average, Linear Regression and Support Vector Regression methods to determine its accuracy and suitability for occupancy prediction. The model will only look at one room for predicting, with limitations of missing measurements, sensor data and the model's ability to predict unexpected changes in the occupancy.

2.0 Input and Output Data

2.1 Input Data

	Meaning	Unit	Allowed Variable
Floor space id	Which Room	Text	Any room number
HeadCount	The total occupancy	integer	Zero or more
CollectedDate	Time of Data	Date and time	Any Date and time

The method uses three variables with the floor space id being filtered out early so that only the selected room can be used to train the model. The Head count is also turned into what is the room occupancy and then the CollectedDate is split into date, hour and minute.

2.2 Output Data

	Meaning	Unit	Intended Use
Input Time	Current Time	Any 15-minute interval time	See the occupancy at that time
Current Occupancy	Current Occupancy at that time	Positive integer	Allows for the model to be compared
Predicted Occupancy	Model's predicted occupancy	Positive integer	Value given to the HVAC system

The main output is the predicted occupancy where it is given in intervals of 15-minutes after the input observation. The Current occupancy is used to evaluate how well the model was able to predict what the actual occupancy will be.

2.3 Data Processing and Limitations

Before the model was made the data was sorted. Firstly, the timestamps were split up and sorted into 15 minute-intervals to be used. Any invalid observations or duplicate timestamps were filtered out this could mean rows with missing head counts. In the situation occurred were there was no new update to the occupancy the previous value is retained up intel 30 minutes where it is then assumed that the room is unoccupied. It allows the dataset to remain continuous without immediately treating missing measurements as zero. However, if the occupancy changes during this period, the retained value may no longer represent the actual conditions of the room. This introduces uncertainty into the model's input data and may reduce the accuracy of its predictions.

3.0 Computational solution

After the data was sorted and organised into its 15-minute intervals it was split up were 70% was training data 15% was validation and 15% was testing data this allows the model not just to learn the data but also be test against unseen data. The method decided was a Decision Tree Regression, a supervised machine-learning technique that is used to predict continuous numerical outputs. The method works by repeatedly dividing the training data into smaller sub-groups using threshold-based decision rules created from the input variables. Each division goal is to reduce the overall variation from the target value; this will then make the final prediction calculated as the mean occupancy of the observation within a terminal node. Do the occupancy of the room showing non-linear relationships to time and previous occupancy and how the change in occupancy takes steps like a decision tree. This method therefore fits the best as it is cable of representing different operating conditions.

The model was done through python the Scikit-learn DecisionTreeRegressor function. The Head count measurements were processed and ran through panda and organised into 15-minute intervals. Six features were created from the core two inputs off occupancy/headcount and time, they were current occupancy, occupancy 15- and 30-minutes prior, minute, hour and day. After this the data was split up were 70% was training data 15% was validation and 15% was testing data this allows the model not just to learn the data but also be test against unseen data. The training data was used to develop the decision rules. The validation set was then used to look at the variables of

the Decision tree with them being the tree- depth and leaf-size which was tested to produce the lowest error. The depth-tree is how many question the function will ask to sort the data. The leaf-size is the minimum number of data-points in each final group. If the tree has to many leaves or is to deep it can result in the function over fitting the data. This is why creating an accurate simple model is the best as it will have a reasonable error while not overfitting.

This approach assumes that the historical occupancy and time-based patterns will be enough to create a model that can successfully predict the occupancy 15 minutes into the future. An additional pre-processing assumption is that if the last time will be retained for 30 minutes however once that 30 minutes is up it is assumed that the new occupancy is 0. This may introduce uncertainty when measurements are missing. Finally, the decision can overfit and struggle to predict unexpected changes in room usage. These issues show the importance of validation and variable optimisation when assessing the suitability of the model for making occupancy predictions.

4.0 Alternative Solutions

Three alternative solutions were investigated to confirm that a Decision regression tree would be the best to predict the room occupancy in 15-minutes. The three methods consisted of the Historical Average, Linear regression and Support Regression Vector (SVR). These methods were selected to compare the model on its predicted accuracy, computational complexity, sustainability for modelling occupancy patterns and implementation complexity.

The historical average was implanted as a baseline comparison for the model as it works by using the mead recorded data for that 15 minute time interval for that day of the week. This method is extremely simple as such the computational complexity is extremely simple however as it runs of an average it means that it is unable to respond unexpected in the room occupancy making the method less accurate at modelling change. Additionally, since only the average is being calculated the memory usage is low and the scalability is extremely easy.

Linear regression was selected as the second alternative mathematical solution, using the six numerical input features to estimate future occupancy. This method is quite computationally efficient and has a low memory usage it struggles to model complex non-linear relationships due to is assuming that the data can be represented by a additive linear relationship reducing the overall accuracy for non-linear relationships. This is important as it helps determine if the Decision Tree was needed to model the non-linear relationship

The third and final alternative solution was the support vector regression (SVR). An RBF kernel was used to model relationships between occupancy and time, with the input variables standardised prior to training. Unlike a Decision Tree which works through doing step like predictions SVR can approximate smooth non-linear relationships. However, SVR require more computational resources and is generally less straightforward to interpret. Additionally, due to it have three parameters its scalability is a little harder as it requires these to be re-tested and adjusted accordingly.

These alternative solutions are important as they evaluate the if the additional flexibility and complexity of the Decision tree was needed. These models are compared on their ability using predictive accuracy and computational performance, while looking at there ability to represent changing occupancy conditions. The quantitative results of these comparisons are presented in Section 6.0, while the optimisation and performance characteristics of the primary Decision Tree model are examined in Section 7.0.

5.0 Validation

The performance of the Decision Tree Regression model was evaluated by comparing its predicted occupancy values against recorded occupancy measurements from an unseen testing dataset. To minimise the information leak between observations, the data was divided into three categories training data which was 70%, testing which was 15% and validation which was 15%. The training dataset was used to develop the model teaching it the relationship between occupancy and the time of day, while the validation dataset is used to compare different model parameters, such as maximum tree depth, and select the settings that produce the lowest prediction error. including maximum tree depth. Then the testing data was used to evaluate the predictions made by each of the models. This split is important to the models as it reduces the risk of the model overfitting the data and provides the ability for it show the accuracy of the data.

Three metrics were selected to measure how well each model was able to predict the occupancy of the room. They are Mean Absolute Error (MAE), Root Mean Squared Error (RMSE) and the Coefficient of Determination (R^2). MAE measures the average absolute difference between predicted and actual occupancy, while RMSE places greater emphasis on larger prediction error, which is particularly important for HVAC systems if the predicted occupancy is significantly different from the actual occupancy. R^2 evaluates the model's ability to account for variation in occupancy relative to a constant mean prediction. Together, these metrics allow the models to be evaluated based on both their average accuracy and their ability to represent changing occupancy conditions.

Models	Test MAE	Test RMSE	R^2
Decision Tree Regression	0.286	0.578	0.452
Historical Average	0.448	0.705	0.185
Linear Regression	0.307	0.590	0.429
Support Vector Regression	0.268	0.590	0.429

From the data we can see that the Support Vector has the lowest MAE against the test data with it being 0.268 and then Decision Tree Regression was second with 0.286. This means that the average absolute error of the SVR was better however a difference of 0.018 people is not that bad when looking at the grand scheme showing that the decision tree was still accurate with its absolute error being small. However, when looking at the RMSE we can see that the Decision Tree was lower than the SVR showing that the decision tree had a lower large variation in the testing and predicted values. As larger errors could lead to inappropriate HVAC responses, RMSE is an important consideration when selecting the most suitable model. When looking at the MAE and RMSE for both historical average and Linear Regression we can see it is higher than the other two methods showing they struggled to represent non-linear patterns and failed to model unpredicted changes in the data. Finally looking at the R^2 we can see that the Decision tree preformed the best with it being the highest of 0.452 whilst the closest was then Linear regression and SVR being 0.429. However, the remaining unexplained variation suggests that the model is unable to fully represent changes in occupancy. This may be due to unexpected changes in room usage, limited input variables or inaccuracies introduced when handling missing data. Although the Decision Tree achieved the highest R^2, further improvements would be required to increase its reliability for practical HVAC applications. Overall, Decision Tree Regression was selected as the preferred model because it achieved the lowest RMSE and highest R^2 while maintaining an MAE close to that of SVR. Even though the SVR provides a better mean error, the decision tree provides a better approach to predicting with its threshold-based decision rules. This makes it easier to understand how historical occupancy and time-based variables influence predictions.

Overall the Decision tree regression performed well. The one thing to look at is the R^2 where it was not bad but also not the greatest indicating that the there is still some variation in the occupancy. The validation was also limited by the dataset's missing measurements and the assumption that occupancy becomes zero after 30 minutes without a new sensor reading. Although this assumption allows for a continuous dataset for training and testing, Consequently, the calculated prediction errors may not

fully represent the model's accuracy when applied to real occupancy measurements. Additionally, with the models being confined to one room it limited how much generalisation can be done to models for other rooms. Overall, the results demonstrate that Decision Tree Regression is a viable approach for short-term occupancy prediction, although further investigation would be required before applying the model to operational building-management systems.

6.0 Optimisation and Performance

Models	Total Runtime (s)	Peak Allocations (Mib)
Decision Tree	0.040s	4.08
SVR	40.474	2.16
Historical Average	0.084	6.52
Linear Regression	0.011	9.06

The first thing done trying to optimise the decision tree was look at how changing the maximum tree depth affected the overall as this controls the complexity of the model and its ability to represent occupancy patterns. A Decision tree with a small depth will result in the patterns not being shown and the occupancy being underfitted while if the depth of the tree is two high it will try to make relationship that don't exist and overfit the data by taking in noise. To look at what is the best depth multiple were tested with the list being 4,6,8,10,12,15 and 20. From this the MAE were compared to each and then the one with the lowest was selected which was 20. The highest depth tested being selected could indicate that there wasn't much noise in the data making the model unable to overfit the data.

The minimum number of observations permitted within each terminal node was fixed at 250 using the min_samples_leaf parameter. This was used to try and reduce the influence a few outliers can have on the overall formation of the rules and questions. Then finally a fixed random state of 42 was chosen to allow the model to be reproduced for other rooms. One way to improve the overall performance of the model would be to test multiple minimum sample leaf's as this could help further reduce the overall MAE.

Following parameter selection, the Decision Tree Regression model achieved a testing MAE of 0.286, an RMSE of 0.578 occupants and an R^2 value of 0.0.452. The model achieved the lowest RMSE and the highest R^2, however the SVR achieved the lowest MAE of 0.268. These results do indicate that the Decision tree was able to best looking model by providing a balance between predictive analysis and model complexity.

The computational performance of the model is important as well to the engineering problem. The complexity of the model can be reduced by reducing the tree depth by reducing the amount of decision to require making a solution. However, when looking at

the overall computational strength of the models were extremely low with the decision tree being 4.08Mibs which is low meaning this aspect is not a problem additionally the runtime is extremely fast.

When looking at how to improve the overall model and is ability to predict the occupancy of the room there is two main ways. Firstly, find a more efficient way to always fill in the wholes of the missing occupancy data as assuming that it will be 0 is not ideal and can lead to the model creating misleading requirements at times. Secondly, train the model against more rooms as this will allow for a model that can be run in multiple buildings not just one room and additionally it will increase the overall sample size allowing the model to be trained off more data potentially increasing the overall accuracy.

7.0 Ai Usage

ChatGPT was used to create scripts of code, componential modelling, troubleshooting and report structure.

I acknowledge my use of ChatGpt to help develop my MMA3001 Numerical Methods and Machine Learning project. Ai used use to assist in the generation of generating and structuring Python code, including data preprocessing, Decision Tree Regression, Linear Regression, Historical Average and Support Vector Regression models. Ai assisted by explain concepts about machine learning, investigation model parameters, resolving programming errors and developing the GitHub repository structure. ChatGpt was additionally used to suggest improvements for the written point.

AI-generated code and material were reviewed by using the python code to verify it produces the correct output and if not relevant changes were made. Additionally the results of this code was inspected by looking at Mean Absolute Error (MAE), Root Mean Squared Error (RMSE) and the Coefficient of Determination (R^2).

Difficulties were faced with the use of AI. With it having errors in the python code, package installation problems, missing dataset columns and inconsistencies between model parameters and documentation. These issued showed that Ai. Is a useful tool but needs to be rapidly tested and the information that it gives cannot immediately assume that the information is create.

My responsibility for this project was chosen if the code created was good along deciding the overall scope of the project, the methods chosen, testing and parameters set. Additionally, the writing and what information is included in the report.

Ai was important to this project as it improved the efficiency in which it was completed as it allowed for python code to be created faster than I could do to my limited knowledge and skills. Additionally, it was important for explaining and furthering my knowledge on machine learning. However, it also shows that Ai needs to be crossed check and tested as the information and code given will contain errors, unsupported assumptions and inconstancies.